

AYMANE AIT DADS

DevOps AI Intern | LLM-Powered Tooling · Anomaly Detection · Intelligent Automation

Ayman.Ait-dads@eurecom.fr | +33 7 60 92 50 93 | linkedin.com/in/aymane-ait-dads | Sophia Antipolis, France

PROFESSIONAL SUMMARY

Built and deployed an end-to-end automated ML pipeline on 100K+ fiber-optic network records at Orange Maroc, reducing manual analysis time by ~60% and delivering AI-driven maintenance recommendations directly to the network optimization team. Engineered an LLM fine-tuning system on a 30B Mamba-Transformer MoE model using LoRA adapters and SFTTrainer, reaching top 10% on a \$106K+ Kaggle leaderboard through 20+ single-variable ablation studies. Brings hands-on skills in Python, prompt engineering, anomaly detection, and LLM orchestration with tool-calling, RAG, and REST APIs aligned to AI-augmented DevOps tooling. Available for internship immediately; based in France with valid student visa - open to USA/Canada placement, visa sponsorship required.

CORE COMPETENCIES

LLM-based troubleshooting | anomaly detection | CI/CD automation | prompt engineering | log analysis | pipeline optimization | NLP model APIs | intelligent monitoring

WORK EXPERIENCE

Orange Maroc

Jul 2024 - Aug 2024

Data Science Intern - Casablanca, Morocco

- Designed and deployed an automated feature engineering and model evaluation pipeline on **100K+ fiber-optic network records**, cutting manual analysis time by ~60% for the network optimization team.
- Built a clustering pipeline using K-Means to map customer performance profiles to **5 commercial service tiers**, with outputs used directly in production network decisions.
- Developed Random Forest classification models on upload speed, latency, and jitter telemetry to detect network quality anomalies across **thousands of nodes**, enabling proactive maintenance.
- Delivered stakeholder presentations translating model outputs into **actionable maintenance recommendations** visualized via Power BI dashboards for non-technical decision-makers.

PROJECTS

LLM Fine-Tuning: NVIDIA Nemotron Reasoning Challenge

Kaggle Competition · In Progress (Jun 2026) · Top 10% Public Leaderboard

- Achieved **top 10% public leaderboard score (0.80/1.0)** on the \$106K+ NVIDIA Nemotron competition by fine-tuning a 30B Mamba-Transformer MoE model with LoRA (r=32, ~888M trainable parameters) via SFTTrainer completion-only loss on 7,828 chain-of-thought reasoning examples.
- Resolved **5 Blackwell GPU (sm_120) incompatibilities** - including monkey-patched allocator warmup, stubbed Mamba/Cutlass imports, and a pure PyTorch RMSNorm fallback - through 20+ single-variable ablations confirming that packing and synthetic data consistently degraded score.

PyTorch · Unsloth · PEFT · Hugging Face Transformers · vLLM · Triton

AI-Powered Anomaly & Log Intelligence Tooling (Anomalous Sound Detection)

EURECOM Industrial ML · 2025 · AUC 0.80

- Built an unsupervised anomaly detection system for industrial slide-rail machinery using a Transformer autoencoder on Mel spectrograms with SpecAugment, achieving **AUC 0.80** for predictive maintenance without any labeled fault examples.
- Designed the full detection pipeline - spectrogram extraction, augmentation, autoencoder training, and reconstruction-error thresholding - demonstrating **pattern detection in system telemetry** directly applicable to log-based anomaly frameworks.

PyTorch · Scikit-learn · NumPy · Matplotlib

EDUCATION

Engineering Degree in Data Science (Master-level) - EURECOM

Sep 2023 - Present

Relevant coursework: Machine Learning, Deep Learning, NLP, Computer Vision, Cloud Computing, Statistics, Web Applications, Image Security

CPGE - Mathematics & Physics - Ibn Timiya

2021 - 2023

SKILLS

AI & LLM Tooling: Python | LLM orchestration | prompt engineering | RAG | tool-calling | LangChain | Claude API

ML Engineering & MLOps: PyTorch | Scikit-learn | Hugging Face Transformers | PEFT | LoRA | anomaly detection | feature engineering

Data & Infrastructure: Pandas | SQL | REST APIs | Docker | PostgreSQL | Power BI | Git | Linux